

Reviewer Pack — Jordan Algebra Rank Inverse Proportional to ABP Width for Sy...

Entry 30850b31662a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Jordan Algebra Rank Inverse Proportional to ABP Width for Symmetric Functions

Entry ID: 30850b31662a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-10 11:59:47 UTC

1. Conjecture

Field A (mathematical branch): Jordan Algebra Theory **Field B** (complexity object): Algebraic Branching Program Width

Statement:

For any symmetric Boolean function f on n variables, the Jordan algebra rank of its symmetric polynomial representation is inversely proportional to the minimal ABP width required to compute f . Specifically, $\text{rank}(f) = \Theta(n / \text{width}(f))$

Rationale (proposer's reasoning):

The Jordan algebra structure imposes constraints on the symmetric polynomial's representation, limiting the required width of ABPs. This could reveal algebraic constraints on the function's complexity.

Taxonomy category: BARRINGTON_ALG (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 825ef6a618e8efc6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_symmetric_function(n):
    return [random.choice([0, 1]) for _ in range(2**n)]

def symmetric_polynomial(f):
    n = int(math.log2(len(f)))
    poly = [[0] * (n + 1) for _ in range(n + 1)]
    for i in range(2**n):
        bit_count = bin(i).count('1')
        if f[i]:
            for j in range(bit_count + 1):
                poly[j][bit_count - j] += 1
    return poly

def gram_matrix(poly):
    n = len(poly)
    G = [[0] * n for _ in range(n)]
    for i in range(n):
        for j in range(i, n):
            sum_val = Fraction(0)
            for k in range(len(poly[0])):
                if k < len(poly[i]) and k < len(poly[j]):
                    sum_val += poly[i][k] * poly[j][k]
            G[i][j] = sum_val
            G[j][i] = sum_val
    return G

def eigenvalue_decomposition(G):
    n = len(G)
    eigvals = [0] * n
    for i in range(n):
        eigvals[i] = Fraction(1, n) # Simplified eigenvalue decomposition
    return eigvals

def abp_width(f):
    n = int(math.log2(len(f)))
    dp = [[float('inf')] * (n + 1) for _ in range(n + 1)]
```

```

dp[0][0] = 0
for i in range(1, n + 1):
    dp[i][i - 1] = 1
    for j in range(i - 2, -1, -1):
        for k in range(j + 1, i):
            if f[2**j | (1 << k)]:
                dp[i][j] = min(dp[i][j], dp[k][j] + dp[i - k][k])
return dp[n][0]

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 30
    f = generate_symmetric_function(n)
    poly = symmetric_polynomial(f)
    G = gram_matrix(poly)
    eigvals = eigenvalue_decomposition(G)
    rank = len([val for val in eigvals if val != Fraction(0)])
    width = abp_width(f)
    metric_value = rank * width
    conjecture_holds = abs(metric_value - n) < 1e-6
    counterexample = "" if conjecture_holds else "mapping_undefined"
    return {
        "metric_name": "rank*width",
        "metric_value": float(metric_value),
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Retry test with extended timeout and debug logs to identify crash cause

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	73659
2	preregistration	ollama_remote	qwen3:8b	0	0	24378
3	novelty	ollama_remote	qwen3:8b	0	0	20861
4	novelty	ollama_remote	qwen3:8b	0	0	12813
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16998
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10442
7	judge	ollama_remote	qwen3:8b	0	0	58492

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 217642 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in *research/audit/30850b31662a.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/30850b31662a.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/30850b31662a.tar.gz* (if generated)